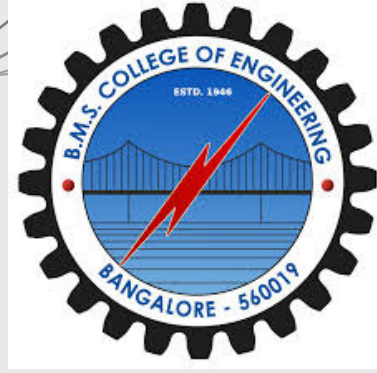




B. M. S. COLLEGE OF ENGINEERING, BENGALURU
DEPARTMENT OF COMPUTER APPLICATIONS

AI-BASED MCA ACADEMIC TIMETABLE SCHEDULER

MINI PROJECT - MMC306

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INTRODUCTION

Academic timetable scheduling is a complex and critical operational task for educational institutions, directly impacting academic efficiency.

Manual scheduling typically results in:

- Class clashes and scheduling conflicts
- Inefficient use of classrooms and labs
- Uneven faculty workload distribution

Furthermore, NEP 2020 introduces added flexibility, significantly increasing complexity and creating a critical need for AI-based automated generation.



PROBLEM STATEMENT

1. Manual and spreadsheet-based scheduling processes are notoriously time-consuming and prone to human error, creating significant administrative bottlenecks in academic institutions.
2. Multiple constraints (faculty availability, room capacity, labs) make conflict-free scheduling difficult.
3. High probability of clashes between faculty schedules, student batches, and classroom allocations.

Goal: To deploy an AI-driven solution that automates the generation of optimized, conflict-free timetables.

EXISTING SYSTEM

Currently, academic timetables are primarily prepared manually using spreadsheets. This traditional approach relies heavily on human intuition and administrative effort to coordinate faculty, students, and resources.

The process is time-consuming, iterative, and prone to human error, often requiring multiple revisions before a finalized schedule is achieved.

Key Limitations:

- High probability of clashes between faculty schedules, batch allocations, or room bookings.
- Poor resource utilization leading to unoccupied labs or classrooms during peak hours.
- Lack of intelligent optimization logic to balance faculty workload evenly.
- Static and non-adaptive nature making mid-semester changes difficult to manage.

PROPOSED SYSTEM

A web-based AI-driven scheduler utilizing constraint-based optimization to automatically generate conflict-free, efficient academic timetables.

- Reduces administrative workload by automating complex scheduling tasks.
- Improves efficiency and planning ensuring optimal resource utilization.
- Scalable and flexible design adapts to changing academic requirements.

HARDWARW AND SOFTWARE REQUIREMENT

Hardware Requirements

Component	Minimum Specification	Recommended Specification
Processor	Intel i3 or equivalent	Intel i5/i7 or equivalent
RAM	4 GB	8 GB or higher
Storage	250 GB HDD	500 GB SSD
Display	1366 × 768 resolution	1920 × 1080 resolution
Network	Stable broadband (2 Mbps)	High-speed broadband (10+ Mbps)

Software Requirements

Category	Tools / Technologies
Frontend	React.js, Bootstrap
Backend	Node.js with Express
Database	MySQL
AI / Optimization	Python (OR-Tools / PuLP)
Hosting	Vercel (Frontend), Render / Railway (Backend)
Operating System	Windows 11 / Ubuntu

SOFTWARE REQUIREMENTS SPECIFICATION

FUNCTIONAL REQUIREMENTS

- ADMIN AUTHENTICATION & ROLE MANAGEMENT
- MANAGE FACULTY, SUBJECTS, ROOMS & CONSTRAINTS
- AUTOMATIC CONFLICT-FREE TIMETABLE GENERATION
- EXPORT TIMETABLE TO PDF/EXCEL FORMATS
- FACULTY FEEDBACK & ADJUSTMENT REQUESTS

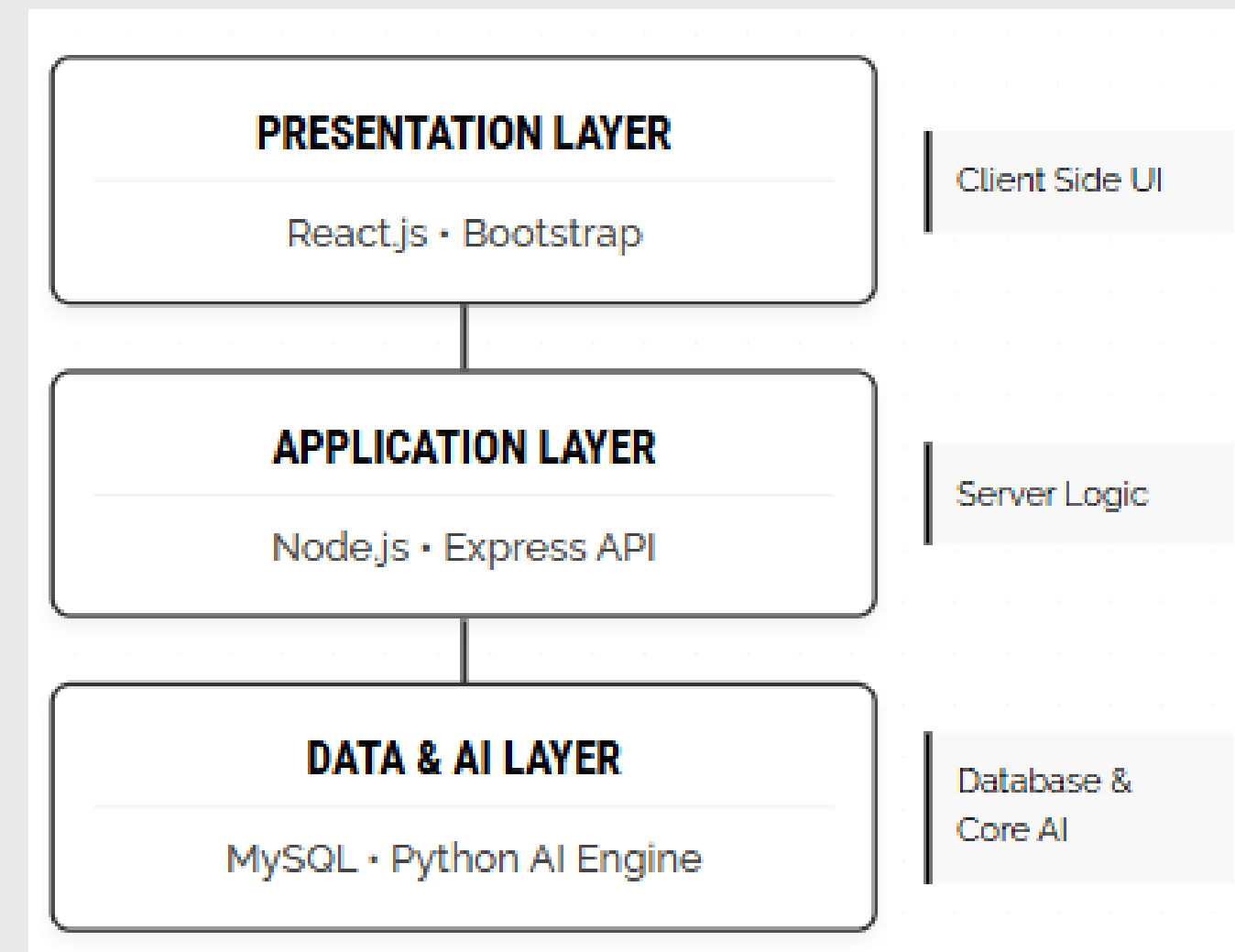
NON-FUNCTIONAL REQUIREMENTS

- USER-FRIENDLY INTERFACE (RESPONSIVE WEB DESIGN)
- EFFICIENT PERFORMANCE (QUICK GENERATION TIME)
- SECURE DATA STORAGE WITH ENCRYPTED PASSWORDS
- SCALABLE TO ACCOMMODATE MORE DEPARTMENTS

SYSTEM ARCHITECTURE

The system follows a robust Three-Tier Architecture, ensuring separation of concerns and maintainability.

- Presentation Layer: Handles user interaction via responsive React.js interface.
- Application Layer: Processes business logic and scheduling requests using Node.js & Express.
- Data & AI Layer: Manages persistent storage (MySQL) and executes complex optimization algorithms (Python AI Engine).
- Design Qualities: Modular, Scalable, Maintainable



DATABASE DESIGN

The system utilizes a normalized relational database schema to efficiently manage the complex relationships between faculty, students, subjects, and infrastructure.

Core Database Tables:

- Users: Authentication & role management
- Faculty :Personal details & availability slots
- Students: Batch, section & semester info
- Subjects: Course codes & lecture hours
- Rooms :Classrooms & lab infrastructure
- Timetable :Generated schedule data
- Feedback: Faculty change requests

APPLICATION MODULES

Admin Module

Centralized control to manage faculty, subjects, and room data.
Responsible for initiating the AI engine to generate and approve final timetables.

Faculty Module

Personalized dashboard for faculty to view their assigned schedules, check workload, and send requests for extra classes or rescheduling.

Student Module

Access point for students to view approved semester timetables, track class locations.

ALGORITHM & WORKING

Constraint Satisfaction: The core engine utilizes a Constraint Satisfaction & Optimization algorithm (e.g., CP-SAT via OR-Tools/PuLP).

Step-by-Step Process

01 : Input Collection: Admin enters subjects, faculty, rooms, hours, and specific constraints.

02: Constraint Handling: System applies logic to avoid faculty overlap, batch clashes, and room conflicts.

03: Optimization: Algorithms run to maximize classroom/lab usage and minimize faculty idle time.

04: Timetable Generation: The engine produces feasible schedule

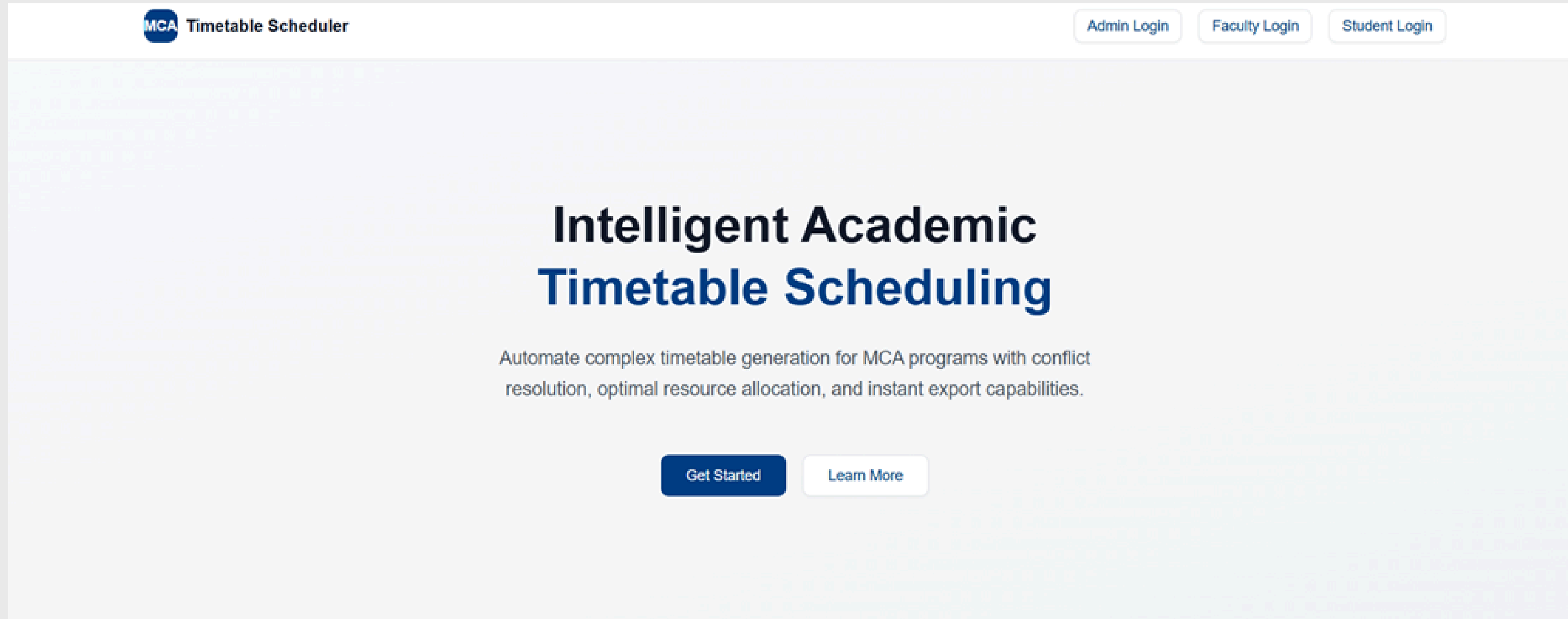
05: Review & Approval: Admin reviews the generated schedule.

06: Conflict Resolution: If conflicts persist, the system suggests alternative slots.

07: Output: Final conflict-free timetable is delivered to students and faculty.

APPLICATION SCREENS

Home Page



This screen shows the home page of the AI Timetable Scheduler application. The page includes navigation options for Admin Login, Faculty Login, and Student Login, allowing role-based access to the system.

Admin module

Manage Academic Data

Configure departments, courses, faculties, and classrooms

[Go to Faculty Assignments](#)[View Timetable](#)[Faculty Requests \(0\)](#)[Generate Timetable](#)[Departments](#)[Courses](#)[Faculties](#)[Classrooms](#)[Students](#)[Add Faculty](#)

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This screen represents the Admin Dashboard of the AI-based timetable scheduler, where administrators manage academic data such as departments, courses, faculty, classrooms, and students.

MCA

Timetable Scheduler

Admin Dashboard

[← Back to Dashboard](#)

Generate Timetable

Create a new timetable version with optimization

Total Classrooms/Labs

7

Courses

18

Faculties

13

Status

Ready

Version Details

Name and describe your timetable version

Timetable Name

e.g., Summer 2024 - MCA Batch A

Description (Optional)

Add notes about this timetable version...

Generate Timetable

Cancel

This screen shows the Generate Timetable page of the AI Timetable Scheduler, accessible to the Admin user. It provides an overview of the current scheduling status, including the number of courses, and faculties available for timetable generation.

Faculty Module

Faculty Dashboard

Welcome, Dr. S uma

[Change Password](#)[Logout](#)

[My Timetable](#)[Send Request \(1\)](#)

[Semester 1](#)[Semester 3](#)

Semester 1

Day / Section	08:55-09:50	09:50-10:45	BREAK	11:15-12:10	12:10-01:05	BREAK	02:00-02:55	02:55-03:50	03:50-04:45
I-A - Monday		MSF KPS CA1	—	PYTHON SU CA1	WEB VPP CA1	—	LINUX 1,2, PY3, DBMS4 SS,GK,RR,KPS Lab 1A		
I-B - Monday		DBMS SS CA2	—	LINUX 4, PY3, DBMS1,2 RMR,TS,SS,KPS Lab 1A		—	WEB VPP CA2		
I-A - Tuesday		DBMS SS CA1	—	LINUX 3, PY1,2, DBMS4 SS,TS,RR,KPS Lab 1A		—	WEB VPP CA1		
I-B - Tuesday	PY1,2, WEB 3,4 SU,TS,VPP,DNS Lab 1A		—	WEB VPP CA2	PYTHON SU CA2	—	DBMS SS CA2		
I-A - Wednesday	LINUX TUTORIAL SS Lab 1A		—	LINUX4, PY3, DBMS1,2 SS,TS,VPP,GK Lab 1A		—	WEB VPP CA1		

This screen displays the timetable, highlighted classes are assigned to a faculty member. Faculty users can view their scheduled classes and send request for extra classes.

Faculty Dashboard

Welcome, Dr. S uma

Change Password

Logout

My Timetable

Send Request (1)

Submit Request

Request Type

Schedule Change

Title

Enter request title

Description

Describe your request in detail

Submit Request

Your Requests

extra class -python
schedule-change
python classe-8:55-9:50 on 29/12/2025
Submitted: 23/12/2025

APPROVED

Faculty can raise requests by selecting the request type, providing a title, and describing the issue in detail. Submitted requests are listed on the same page for tracking their status.

Student Module

MCA Student Timetable									
Archana Logout									
Semester 3									
Day / Section	08:55-09:50	09:50-10:45	BREAK	11:15-12:10	12:10-01:05	BREAK	02:00-02:55	02:55-03:50	03:50-04:45
II-A - Monday	ADA 1,2 ,DT 3,4 TSP,PLN,DNS,VK 📍 Lab1A		—	ELE-3 GK,TSP,RR 📍 CA2,CA3,L1B	CC RR 📍 CA2	—	STP TS 📍 CA1	ELE-2(AI,CS) VK,DNS 📍 CA1,CA3	
II-B - Monday		CC TS 📍 CA3	—	ELE-3 GK,TSP,RR 📍 CA2,CA3,L1B	STP PLN 📍 CA3	—	ADA TSP 📍 CA3	ELE-2(AI,CS) VK,DNS 📍 CA1,CA3	
II-A - Tuesday	ELE-2 DevOps RMR 📍 L1B	ELE-3 GK,TSP,RR 📍 CA2,CA3,L1B	—	Agile DNS 📍 CA3	ADA TSP 📍 CA3	—	STP TS 📍 CA3	ELE-3 LAB GK,TSP,RR 📍 FDC,L1B,L2	
II-B - Tuesday	ELE-2 DevOps RMR 📍 L1B	ELE-3 GK,TSP,RR 📍 CA2,CA3,L1B	—	DT Tutorial GK 📍 CA1		—	ADA 3,4 DT 1,2 PLN,KPS,DNS,VK 📍 Lab1A		
II-A - Wednesday	DT TUTORIAL GK 📍 CA1		—	ELE-2 VK,DNS,RMR 📍 CA1,CA3,L1B	ADA TSP 📍 CA3	—	Agile Tutorial DNS 📍 CA1		
II-B - Wednesday	CC TS 📍 CA3	ADA TSP 📍 CA3	—	ELE-2 VK,DNS,RMR 📍 CA1,CA3,L1B	STP PLN 📍 CA1	—	Agile Tutorial VK 📍 CA3		

This page allows students to view the finalized and approved timetable. Students can easily check class schedules

TESTING AND VALIDATION

Comprehensive testing was performed to ensure the accuracy, reliability, performance, and security of the AI-Based Timetable Scheduler. Multiple testing levels were conducted to validate both individual components and the complete system under real-world conditions.

System Testing : System testing evaluated the application as a fully integrated system..

Functional Testing

- Verified end-to-end timetable generation
- Ensured conflict-free scheduling (faculty, rooms, labs)
- Validated role-based access for admin, faculty, and students
- Confirmed correct storage and retrieval of timetables

Non-Functional Testing

- Performance Testing: Tested system responsiveness with large datasets and multiple requests
- Security Testing: Verified authentication, authorization, and role-based access control
- Usability Testing: Ensured user-friendly interface and smooth interaction for all users

All critical testing phases were successfully completed. The system consistently generated **conflict-free** and **optimized** timetables, maintained acceptable performance under load, and enforced secure access control. Overall, the system is stable, reliable, and ready for deployment in an academic environment.

Functional Testing

Test Case ID	Description	Input / Action	Expected Output	Actual Output	Status
FT-01	End-to-end timetable generation	Enter data and trigger generation	Valid timetable generated	Generated successfully	Pass
FT-02	Conflict-free schedule validation	Analyze generated timetable	No faculty or room conflicts	No conflicts found	Pass
FT-03	Role-based access	Login as admin, faculty, student	Access based on role	Correct access granted	Pass
FT-04	Timetable storage	Save generated timetable	Timetable stored in database	Stored successfully	Pass
FT-05	Timetable display	View timetable	Correct timetable displayed	Time table Coming Soon	Fail

Non-Functional Testing

Test Case ID	Test Type	Description	Input / Action	Expected Output	Actual Output	Status
NFT-P-01	Performance	Timetable generation under load	Large dataset with increased subjects, faculty, and constraints	Timetable generated within acceptable response time	Generated within limits	Pass
NFT-P-02	Performance	Multiple timetable generation requests	Repeated generation requests in short duration	Stable performance with consistent output	Same timetable generated repeatedly	Fail
NFT-S-01	Security	Unauthorized access attempt	Access admin page without authentication	Access denied	Access denied	Pass
NFT-S-02	Security	Role-based access restriction	Student attempts to access faculty features	Access restricted based on role	Student logged in successfully	Fail



CONCLUSION

- Successfully automated the generation of academic timetables.
- Eliminated manual scheduling errors and reduced conflicts.
- Improved overall administrative efficiency and academic planning.
- Demonstrated practical application of AI in academic resource management.
- System is scalable and ready for real-world deployment.



FUTURE ENHANCEMENTS

- **LMS Integration**

Seamless synchronization with Learning Management Systems to automatically update student and faculty dashboards with schedule changes.

- **Real-time Faculty Availability**

Live tracking integration to handle unexpected faculty leaves or availability changes for dynamic, on-the-fly rescheduling.

- **Mobile Application**

Dedicated iOS and Android mobile apps providing instant access to timetables, push notifications, and change requests.

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- **MySQL Documentation** :MySQL 8.0 Reference Manual. *<https://dev.mysql.com>*
- **Node.js & Express** Official API Reference. *<https://nodejs.org>*
- **React.js** Official Frontend Documentation. *<https://reactjs.org>*



THANK YOU!!

Questions and Feedback Welcome

